Introduction to periodontics

Periodontics

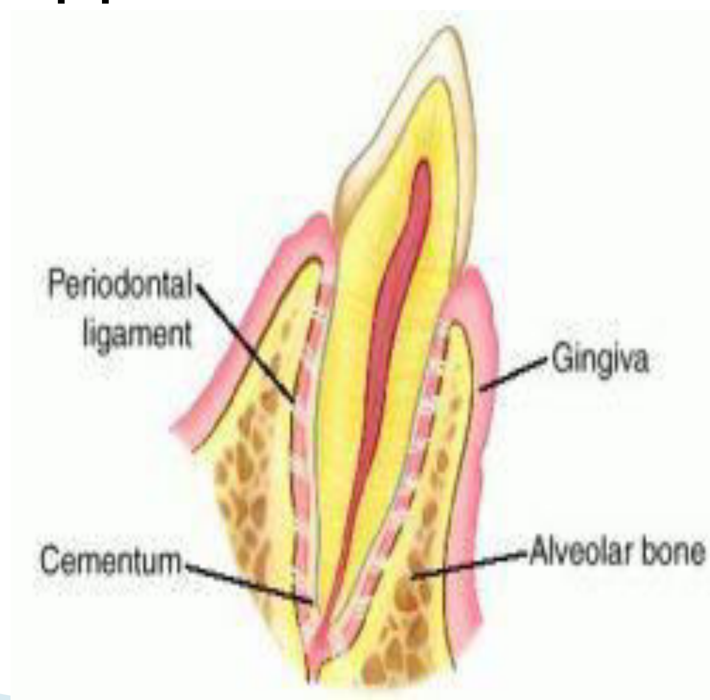
- Area of dentistry that focuses on tissues that surround and support the teeth.

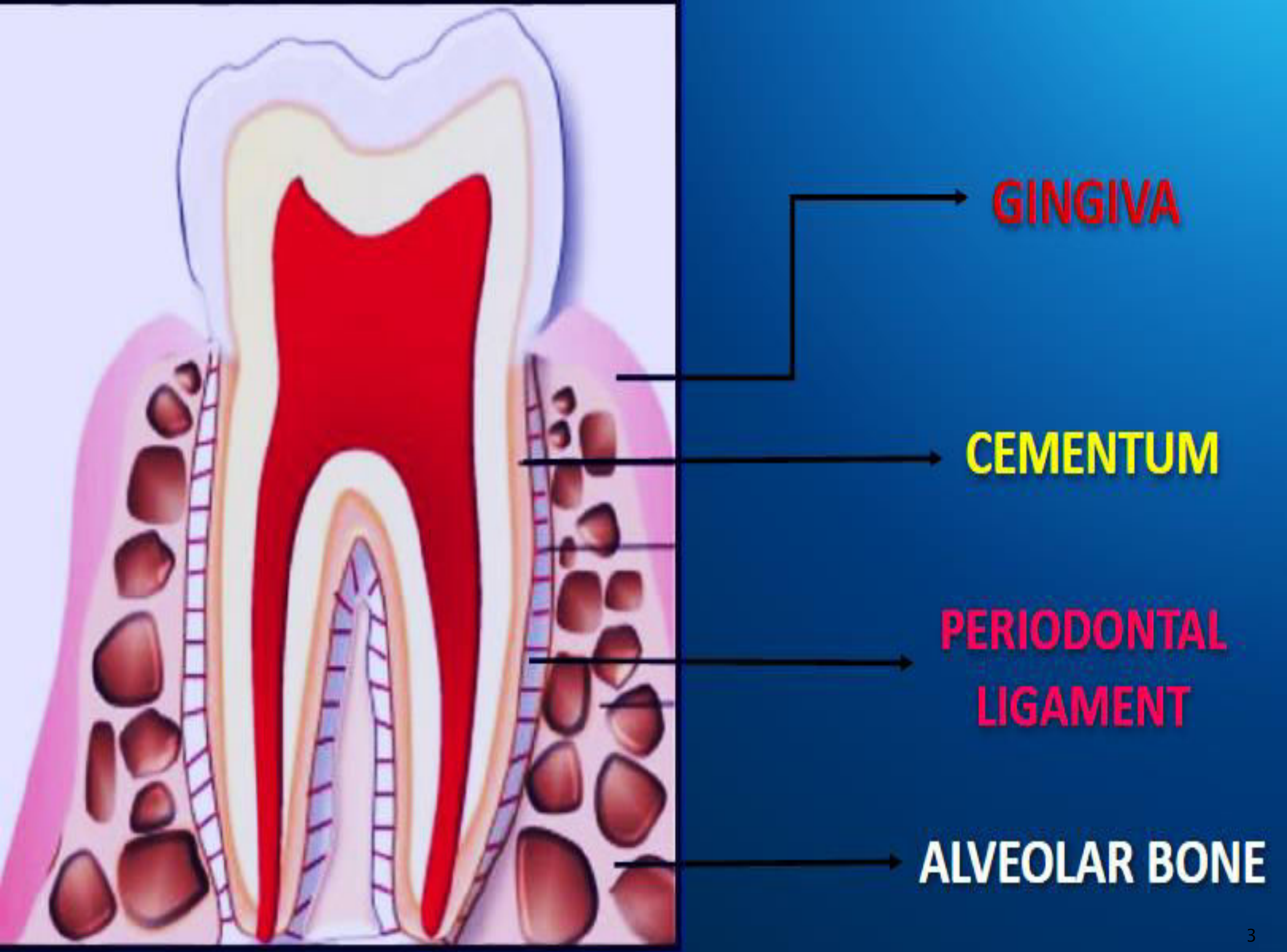
Periodontium:

The tissues that surround, support and are attached to the teeth .

that consists of :

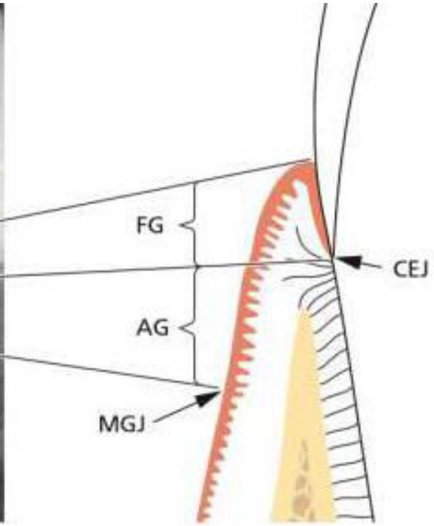
1) Gingiva .
2) Alveolar bone.
3) Periodontal ligaments.
4) Cementum .

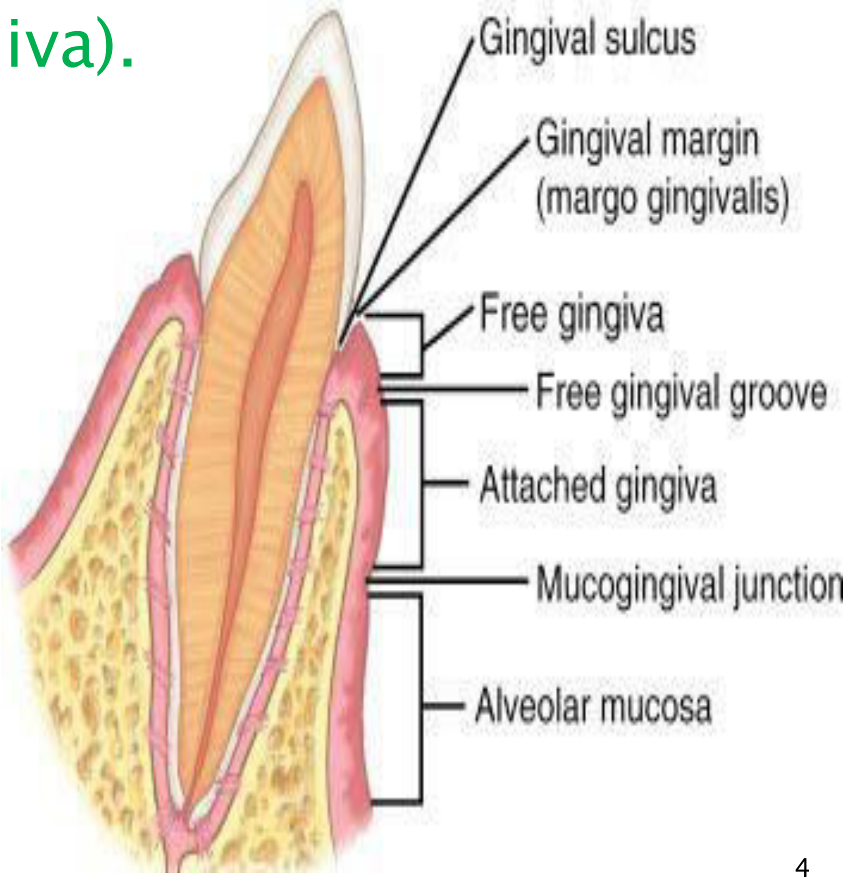



GINGIVA
CEMENTUM
PERIODONTAL
LIGAMENT
ALVEOLAR BONE

Gingiva

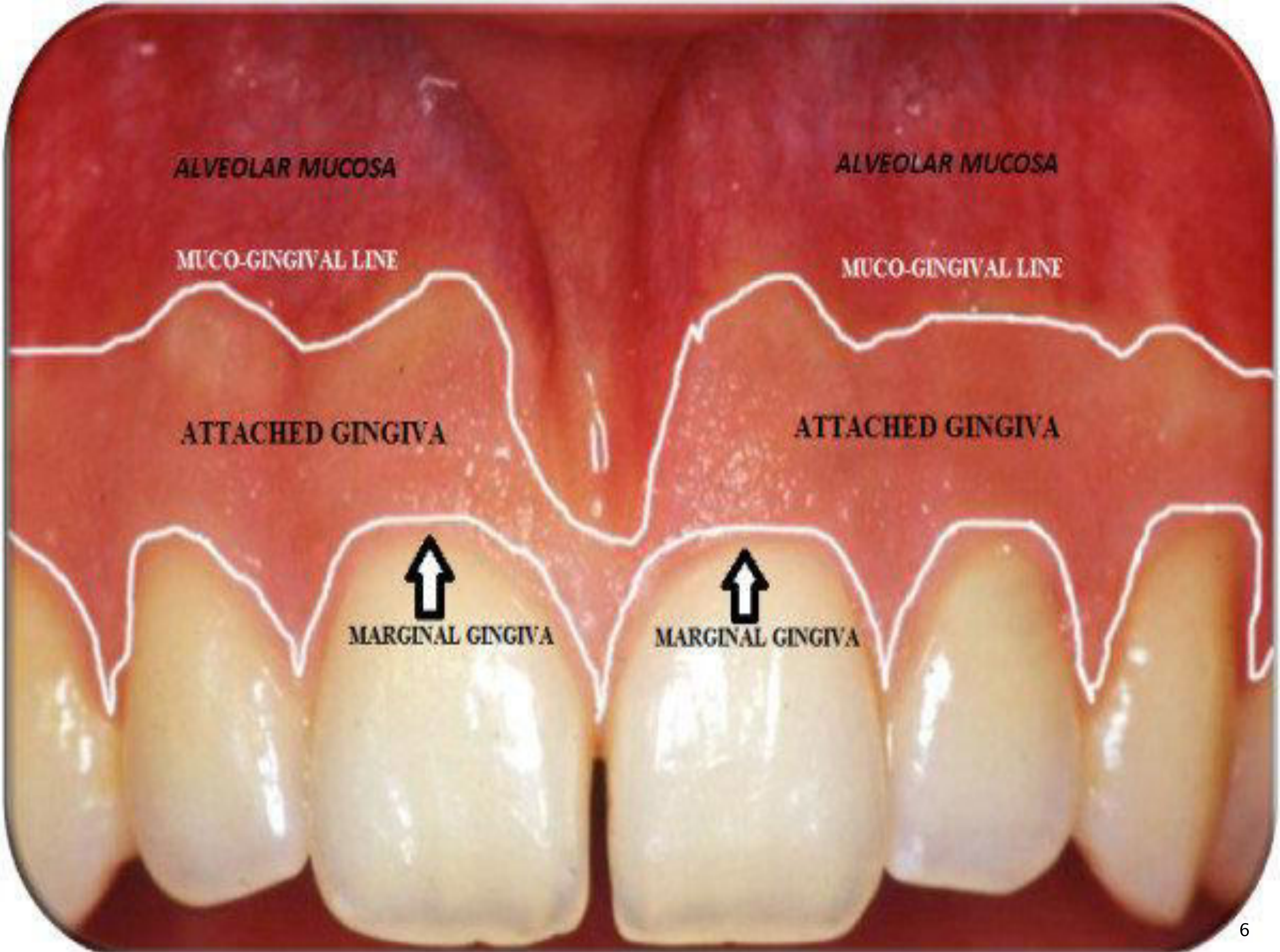
- Covers the alveolar bone and surrounds the neck of the teeth. is the only visible part of periodontium.
- **The gingiva is divided anatomically into:**
  - Marginal gingiva (free gingiva).
  - Attached gingiva .
  - Interdental gingiva .





ALVEOLAR MUCOSA

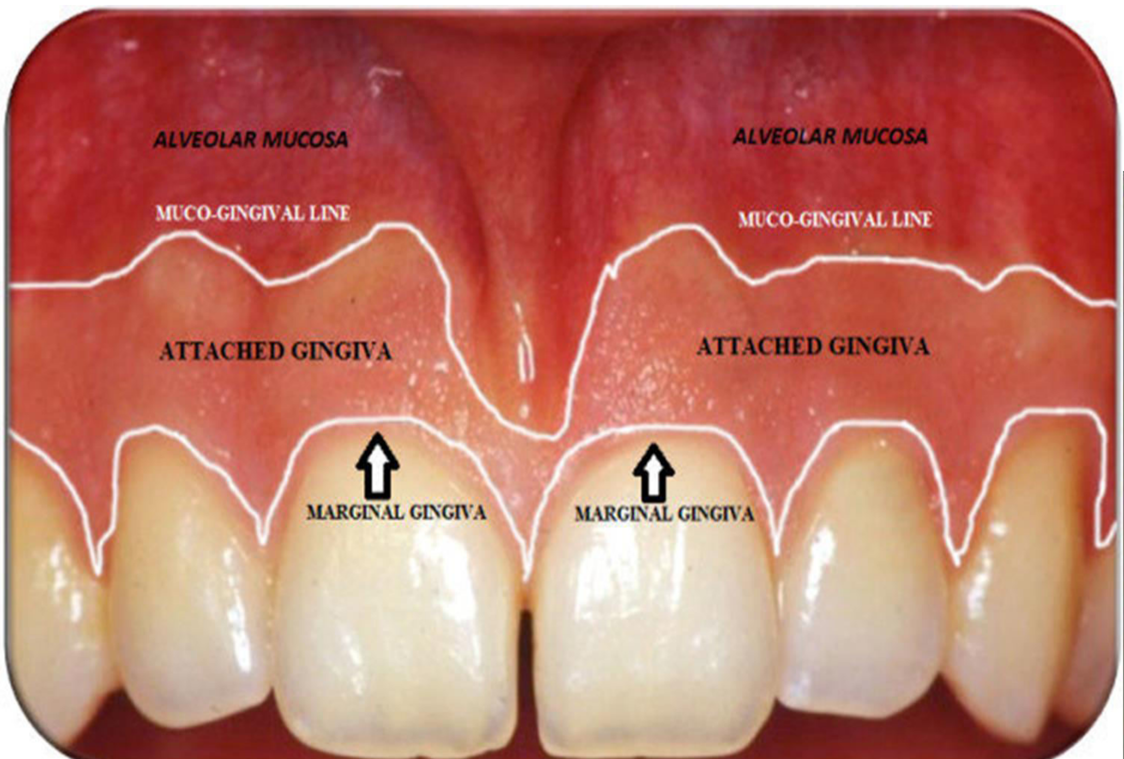
- NOT PART OF THE PERIODONTIUM
- THIN, RED, SMOOTH TISSUE
- MOVABLE


ALVEOLAR MUCOSA
ALVEOLAR MUCOSA
MUCO-GINGIVAL LINE
MUCO-GINGIVAL LINE
ATTACHED GINGIVA
ATTACHED GINGIVA
MARGINAL GINGIVA
MARGINAL GINGIVA

Marginal gingiva

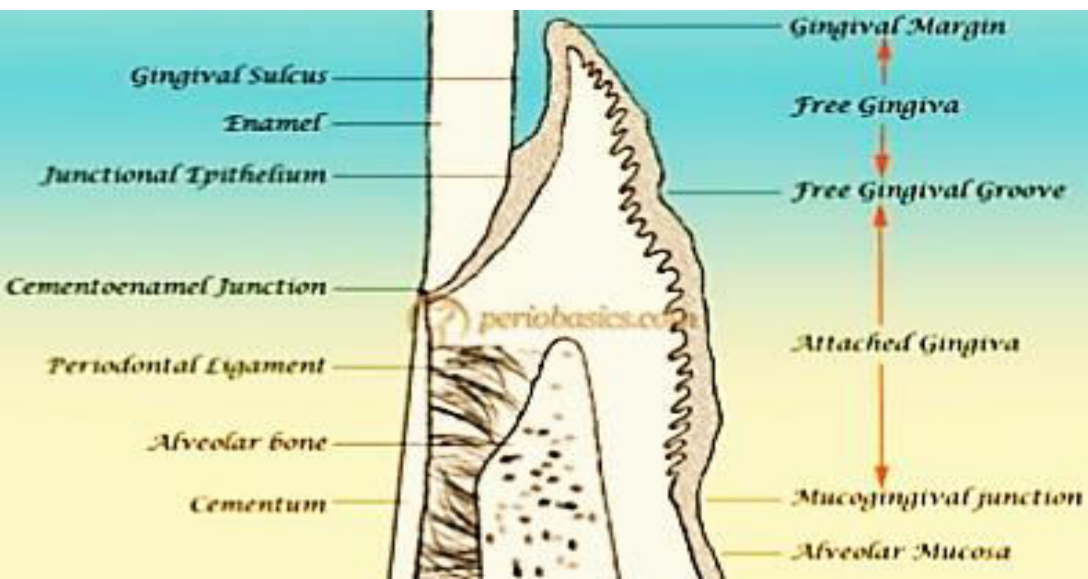
- The marginal , or free or unattached gingiva is the terminal edge or border of the gingiva surrounding the teeth

- The movable portion that is not directly attached to the bone or tooth surface. So Can be separated from the tooth by a probe



Marginal gingiva (cont.)

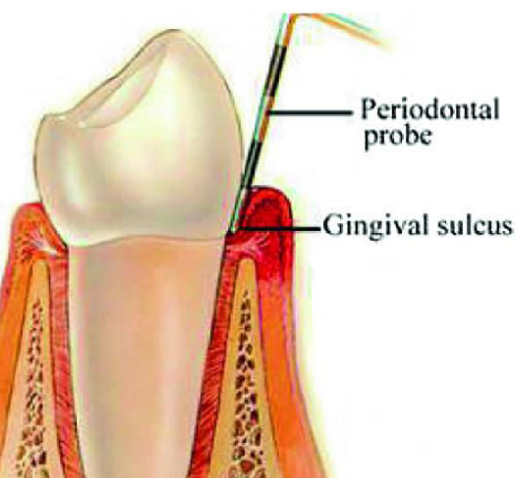
- 30-40 % cases- it is demarcated or separated from the adjacent attached gingiva by a shallow linear depression called the **free gingival groove**

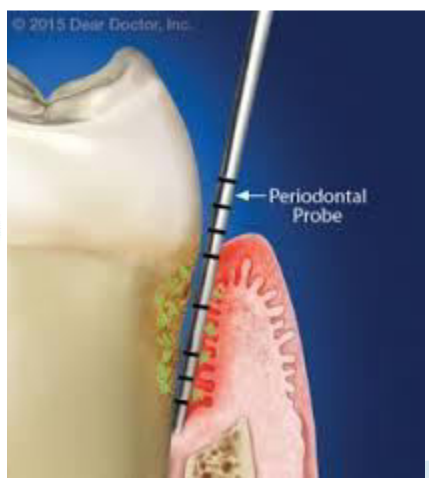


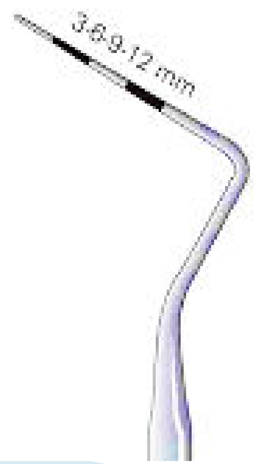
Gingival sulcus:

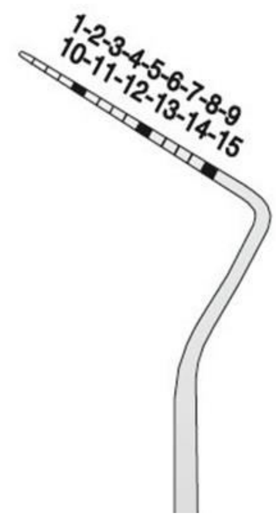
- A shallow crevice(space or groove) around the tooth between the tooth & surrounding free gingiva.
- V- shaped
- Normal depth of **Gingival Sulcus** is 1-3 mm, while Pathological periodontal pocket depth is more than 3mm.
- The clinical determination of the depth of the **Gingival Sulcus** is an important diagnosis method.

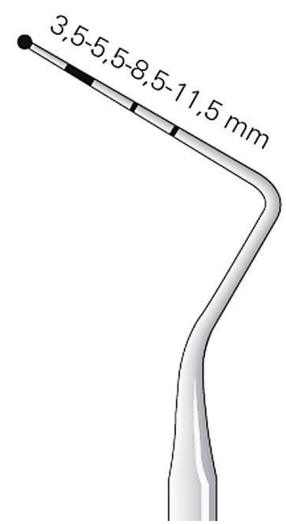
Periodontal probe: a device used to measure the pocket depth & for examination.











Interdental gingiva

- Fills the area between the teeth.
- It’s a part of free gingiva.
- The color of the gingiva is generally described as “coral pink"
- With Healthy gingiva no redness, no swelling, no bleeding on probing.

Attached gingiva

- Continuous with marginal gingiva & is firm,tightly bound to the underlying alveolar bone and cementum .
- Non movable.

Periodontal ligament (PDL)

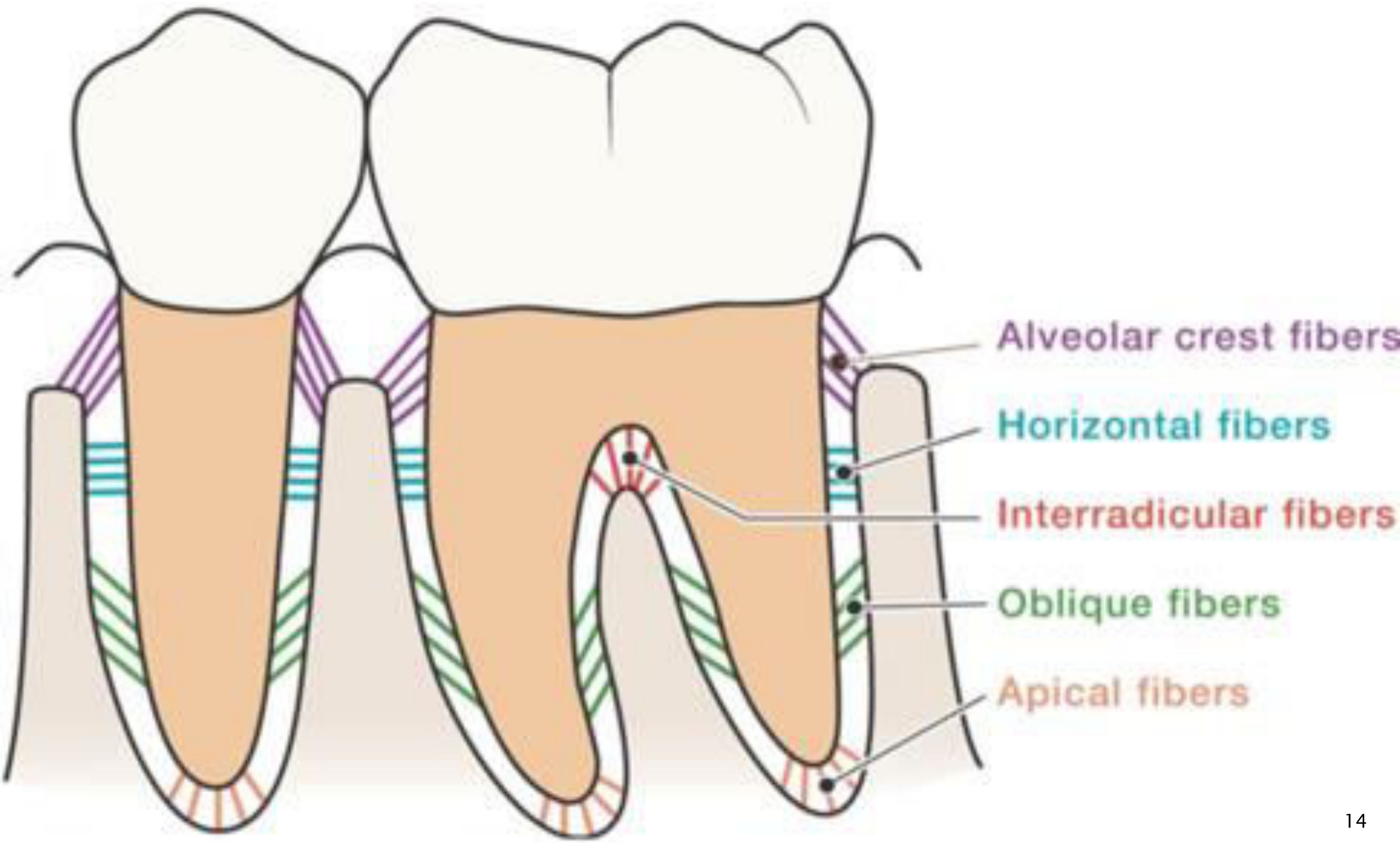
- A group of specialized connective tissue fibers that attach a tooth to the alveolar bone.
- Fibers are arranged in bundles.
- Forms a shock-absorber for the tooth in the socket.
- Composed of :

1) Fibers for attachment .
2) Cells .
3) Sensory Nerves .
4) Blood & lymphatic vessels .

Periodontal ligament functions:

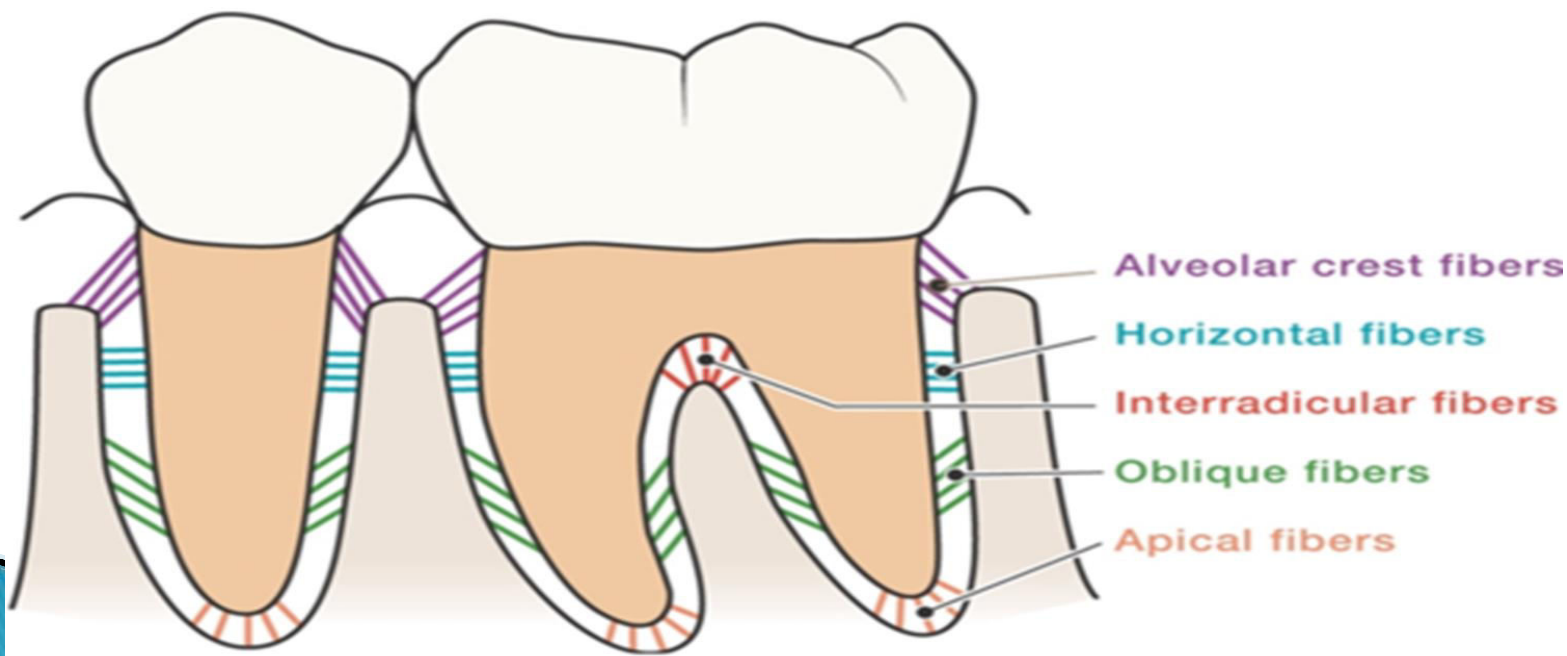
- Supportive: maintains tooth in socket.
- Sensory: sense of touch,pressure,pain.
- Formative: fibroblasts,cementoblast,osteoblasts.
- Protective: cushion-like action from shock.
- Nutritive- blood vessels provide nutrients.

Principal fiber groups of the periodontal ligament



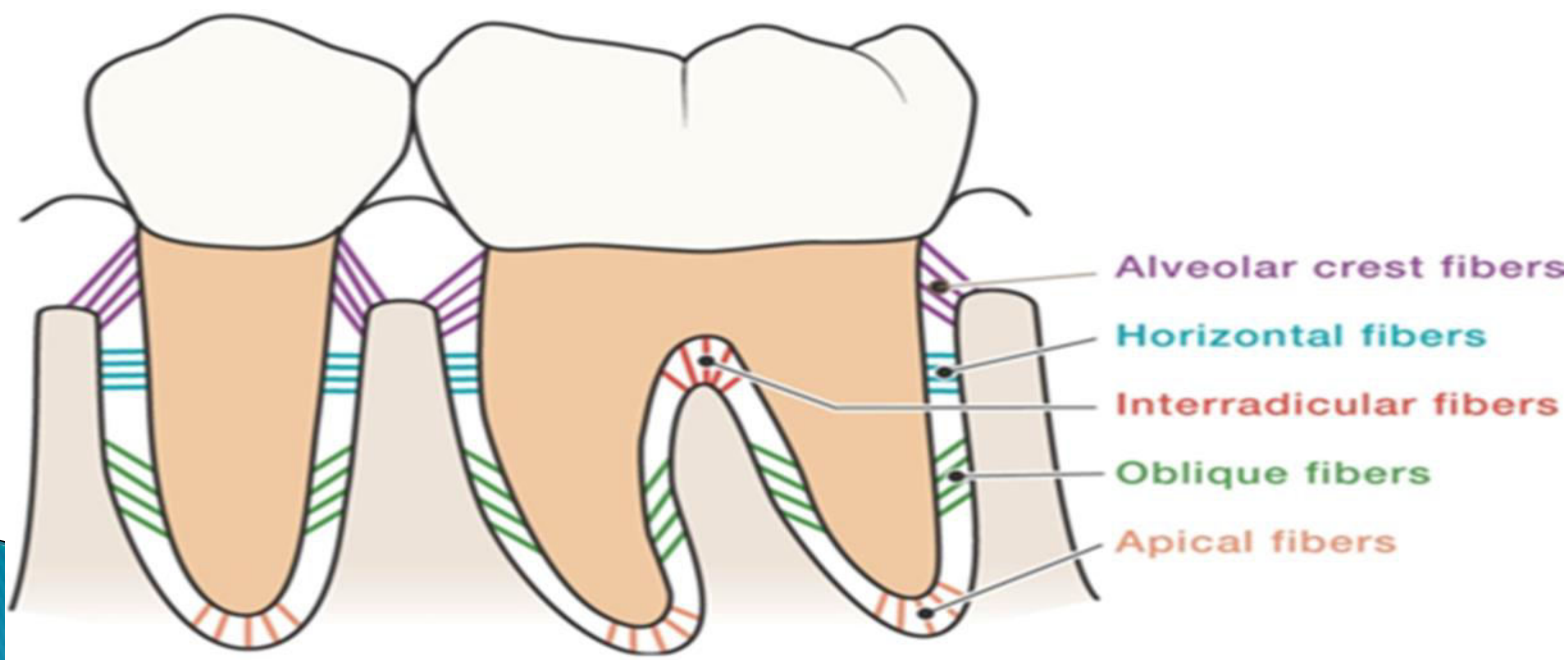
Alveolar Crest Fibers

- Runs apically from cementum to insert into the alveolar crest.
- The first fibers to be formed.



Horizontal Fibers

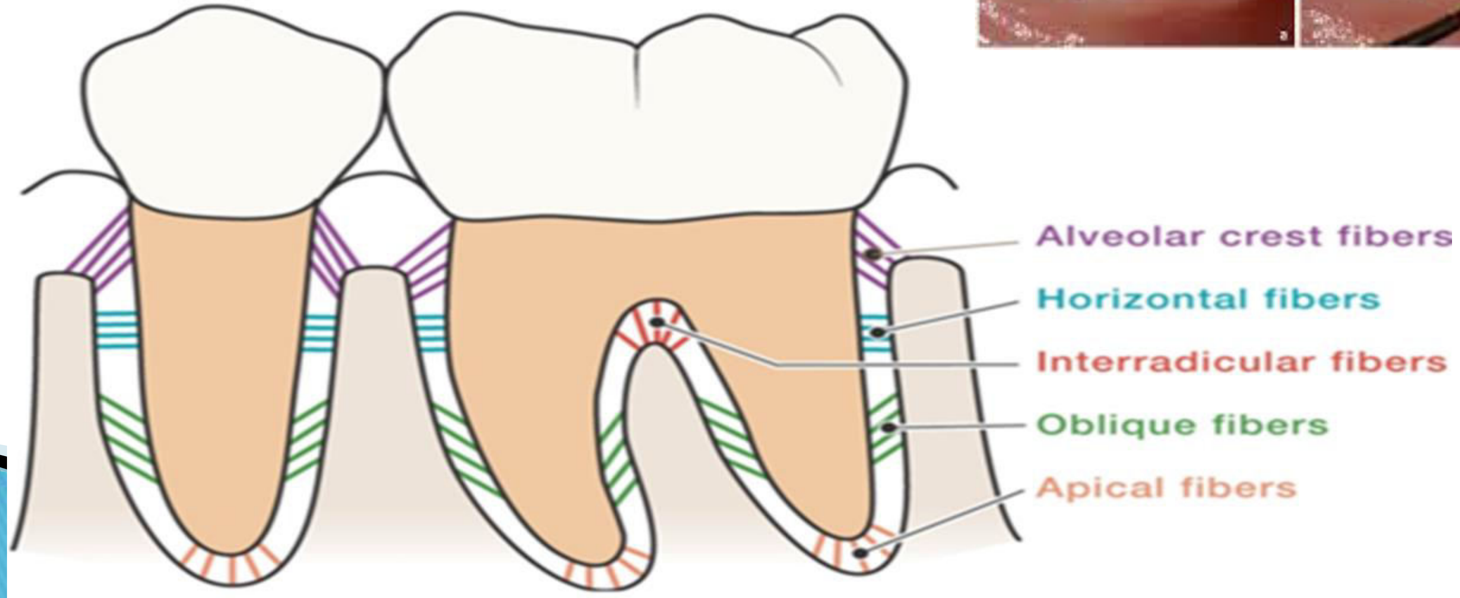
- Runs from cementum to the bone at right angles.
- The second fiber to be formed.



Interradicular Fibers

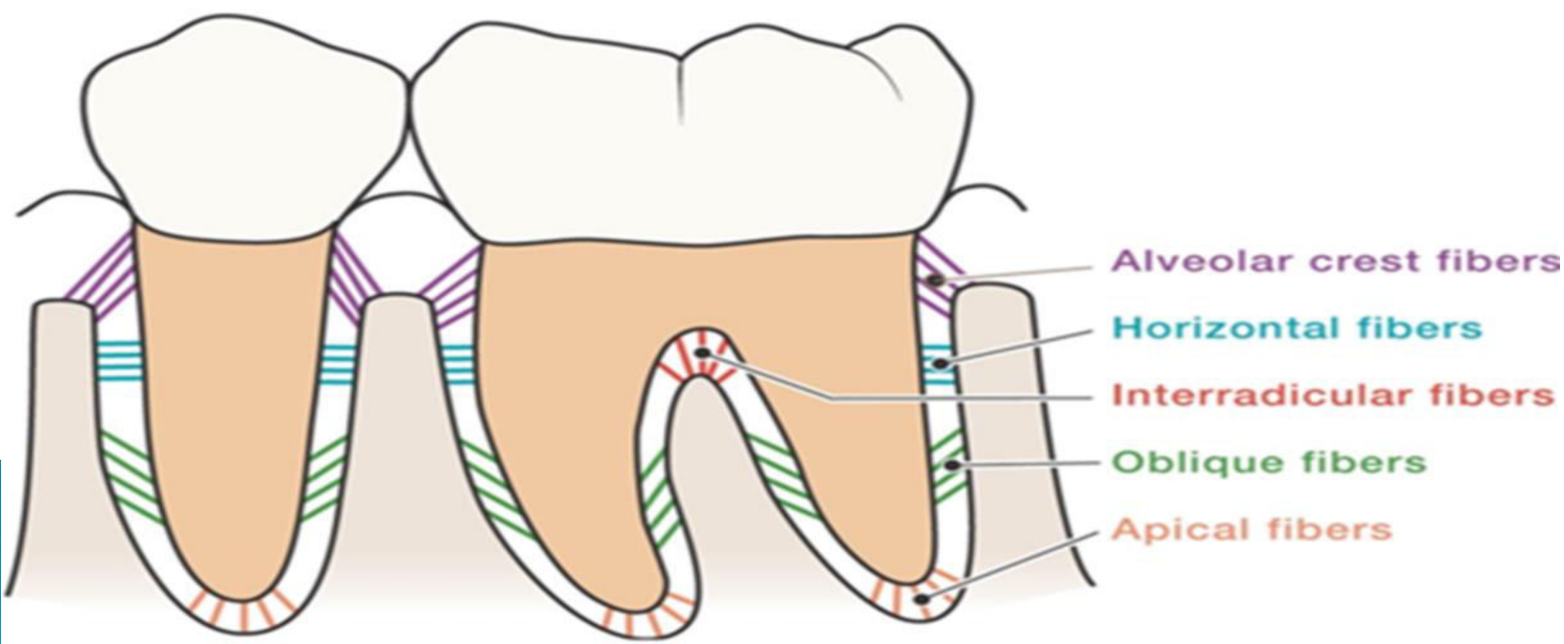
- Only found between the roots of multirooted teeth such as molars.
- From cementum to nearby alveolar bone.

- **Furcation area**: the anatomical area on a multirooted tooth where the root divides into 2 or 3 divisions.



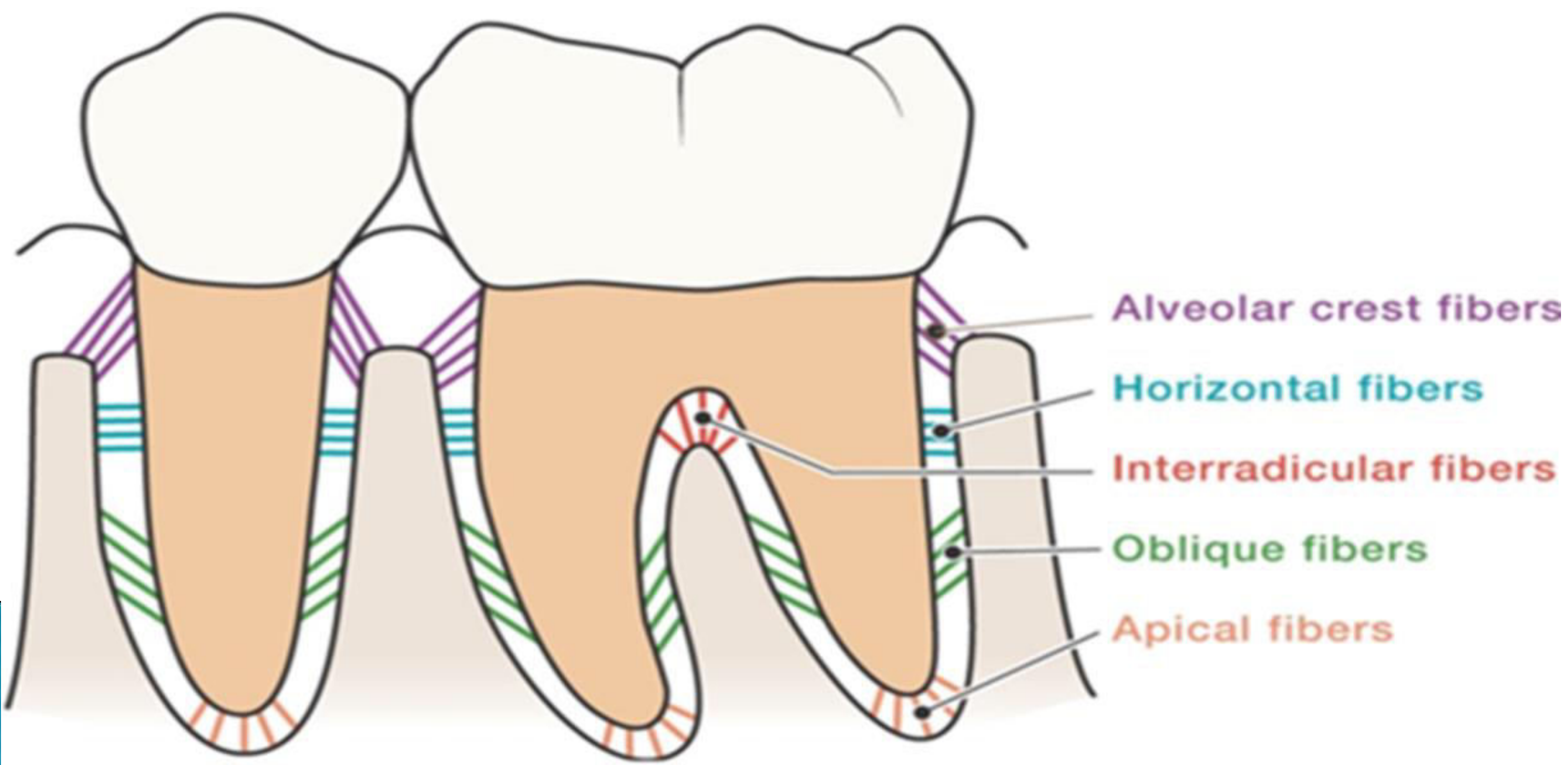
Apical Fibers

- Originates from cementum of the apex of the root, spreading out apically & laterally into bone.
- One of the last fibers to form.



Oblique Fibers

- Located apical to the horizontal fibers; originates from cementum & runs coronally to the bone.
- The most abundant(most numerous).



Periodontal disease

Primary cause of tooth loss in adults

Gingivitis

- **Dental plaque** is soft biofilm that accumulates naturally on the teeth surfaces. Microorganisms is the most numerous component of plaque which is the primary cause of gingivitis.
- Constant irritation to the gum leads to inflammatory process. in early stage called gingivitis.
- **Gingivitis** does not affect the underlying supporting structures of the teeth (e.g. alveolar bone, PDL….)

Signs of gingivitis include:

- Red gum.
- Puffy(swollen) gums
- Bleeding after brushing.
- Painful to touch.
- Bad breath(halitosis)

Gingival Bleeding on probing

- Bleeding on probing is easily detectable clinically and is of value for the early diagnosis and prevention of more advanced gingivitis.

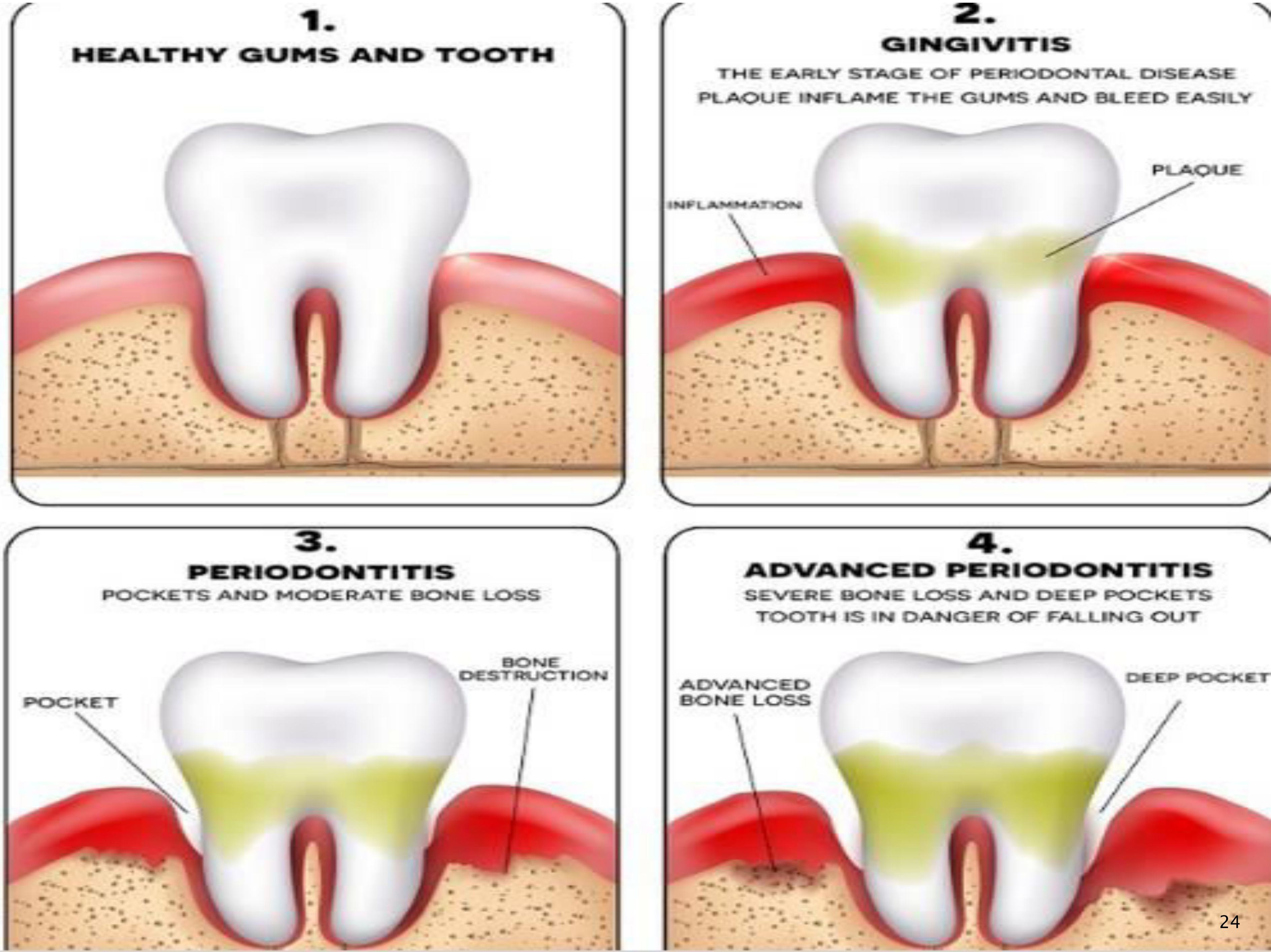
- Earliest sign: appears earlier than change in color or other visual signs of inflammation

- Gingivitis is Reversible to health once the causative plaque (biofilm) is removed.

- But if left untreated, inflammation will spread from gum deep into the periodontal ligaments.

- **Periodontitis** which is a destructive form of periodontal disease, with attachment loss and bone loss with periodontal pocket formation, gingival recession, or both

- Is a major cause of tooth loss in adults


1.
HEALTHY GUMS AND TOOTH
2.
GINGIVITIS
THE EARLY STAGE OF PERIODONTAL DISEASE
PLAQUE INFLAME THE GUMS AND BLEED EASILY
INFLAMMATION
PLAQUE
3.
PERIODONTITIS
POCKETS AND MODERATE BONE LOSS
POCKET
BONE DESTRUCTION
4.
ADVANCED PERIODONTITIS
SEVERE BONE LOSS AND DEEP POCKETS
TOOTH IS IN DANGER OF FALLING OUT
ADVANCED BONE LOSS
DEEP POCKET

Dental Calculus

- Mineralized dental plaque that forms on the surfaces of natural teeth and dental prosthesis also known as tartar.

- The soft plaque is hardened by the precipitation of mineral salts to form hard calculus.

- Calcifying plaques may become 50% mineralized in 2 days and 60% to 90% mineralized in 12 days.

Dental calculus (cont.)

- Dental calculus is **classified by its location** in relation to the adjacent free gingival margin:

Supragingival calculus: located **coronal** to the margin of the gingiva and **visible** in the oral cavity.

Most common locations:

mandibular anterior teeth

maxillary molars

- **Subgingival calculus:** is located below the crest of the marginal gingiva, usually in periodontal pockets, not visible upon oral examination.

- Occurs with or without supragingival calculus.

- The process of calculus cleaning is called scaling. by scaler.
- Dental plaque cleaning by tooth brush.



Risk factors for gingivitis

- **Plaque-host interaction can be altered by the effects of local factors, systemic factors, medications, and malnutrition**

- **Smoking :** regular smokers more commonly develop gingivitis, compared with non-smokers.
- **Pregnancy :** hormonal changes as progesterone hormone & estrogen.
- **Prosthodontic treatment**: bridges & crowns.
- **Drugs :** cause gingival enlargement as anticonvulsants drugs : phenytoin
  immunosuppressant drugs : cyclosporine
- **Mouth breathing..**

- **Diabetes Mellitus**

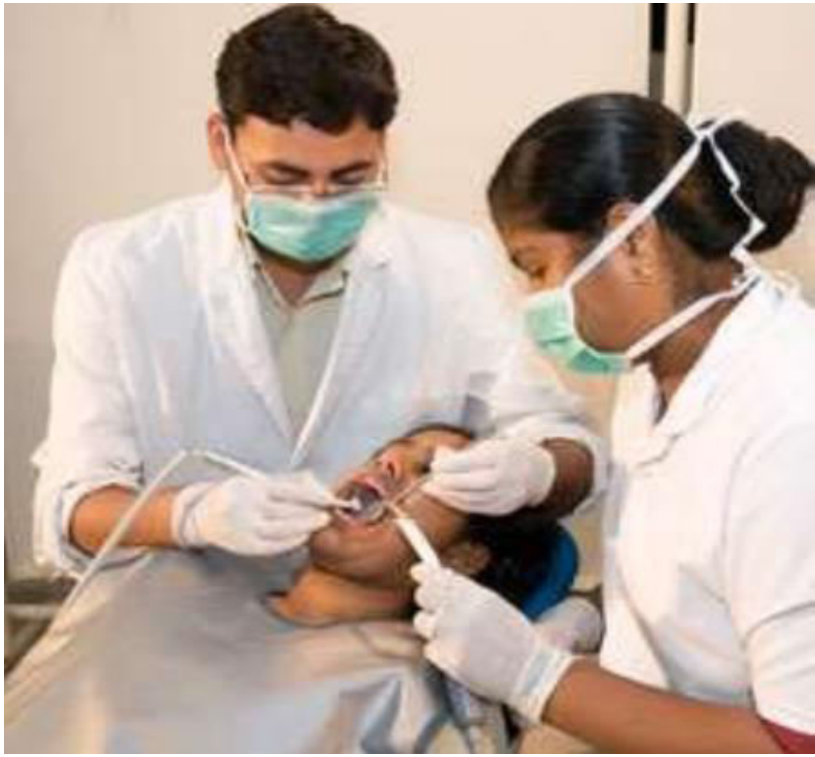
Prevention of gingivitis

- Teeth brushing at least twice a day.
- Teeth flossing at least once a day
- Regularly rinse mouth with an antiseptic mouthwash(chlorhexidine).

The number of bacteria in the oral cavity is greater in the morning –(low saliva during sleeping time)- So, it is very important to brush your teeth before you go to sleep

Treatment of periodontal disease

- Oral hygiene instructions.
- Periodic examination and regular dental visits
- Scaling
- Periodontal Surgery.
- Medications….



THANK YOU